

Clauses

Clause order: SELECT → FROM → WHERE → GROUP BY → HAVING
→ ORDER BY → LIMIT

SELECT x	Selects the column x from the table.
SELECT *	Selects all columns from the table.
SELECT DISTINCT x	Selects unique values from the column x .
FROM table	Specifies the table to select from.
WHERE x	Filters the results based on the condition x .
ORDER BY x	Sorts the results based on the column x .
GROUP BY x	Groups the results based on the column x .
AS x	Renames a column or table to x .
HAVING x	Filters the results of a GROUP BY clause based on the condition x .
LIKE x	Used in WHERE clauses to match a pattern. "%" many characters, "_" single character
LIMIT x	Limits the number of rows returned to x .

Operator

LIKE x	Used in WHERE clauses to match a pattern. % represents zero, one, or multiple characters. _ represents a single character.
AND x	Combines multiple conditions in a WHERE clause.
OR x	Combines multiple conditions in a WHERE clause.
NOT x	Negates a condition in a WHERE clause.
IN (x , y)	Checks if a value is in a list of values.
BETWEEN x AND y	Checks if a value is within a range of values.
IS NULL	Checks if a value is NULL.
IS NOT NULL	Checks if a value is not NULL.

Functions

TOP(n)	Limits the number of rows returned to n .
COUNT(x)	Counts the number of non-null values in column x .
SUM(x)	Calculates the sum of values in column x .
AVG(x)	Calculates the average of values in column x .
MIN(x)	Finds the minimum value in column x .
MAX(x)	Finds the maximum value in column x .

Examples

```
SELECT Orders.OrderID, Customers.FirstName,  
Customers.LastName, Customers.Email, Orders.OrderDate  
FROM Orders  
JOIN Customers ON Orders.CustomerID=Customers.CustomerID  
ORDER BY OrderID
```

Nested Queries

```
SELECT * FROM Orders  
WHERE CustomerID=(  
    SELECT CustomerID FROM Customers  
    WHERE Email="ex@example.com"  
)
```